MODERN TOPICS IN INFORMATION TECHNOLOGY

Assignment Report

Assignment 02

Bachelor of Science Special (honors)

In Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2022

DECLERATION

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any

other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	**Student ID**	**Function Name**
Priyadarshana K.W.P.S.D	IT18007398	Eiffel Tower
Deshitha Thilindra H.B	IT19049496	Moai Statues
Sandun Harshana S.G.	IT19055954	Maison Carrée
Kavinda K.T.S	IT18146820	Giza Pyramid

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

…… ……

Signature of the supervisor Date

Contents

DECLERATION 2

INTRODUCTION 5

ASSIGNMENT TASKS 5

METHEDOLOGY 5

IMPLEMENTATION 6

Function 1-Eiffel Tower 6

Step 1 – Creating a secure server 6

Step 2 – Creating a project 7

Step 3 – Run the server 8

Step 4 – Open the browser and browse on the URL 9

..... **Error! Bookmark not defined.**

CODE EXPLAINATION 10

Function 2- Moai Statues 11

Step 1 – Creating a secure server 11

Step 2 – Creating a project 12

Step 3 – Run the server 13

Step 4 – Open the browser and browse on the URL 15

..... **Error! Bookmark not defined.**

CODE EXPLAINATION 16

Function 3- Maison Carrée 18

Step 1 – Creating a secure server 18

Step 2 – Creating a project 18

Step 3 – Run the server 19

Step 4 – Open the browser and browse on the URL 21

..... **Error! Bookmark not defined.**

CODE EXPLAINATION 22

Function 4 – Giza Pyramid 23

Step 1 – Creating a secure server 23

Step 2 – Creating a project 24

Step 3 – Run the server 25

Step 4 – Open the browser and browse on the URL 26

..... **Error! Bookmark not defined.**
CODE EXPLAINATION 27
FINAL APPLICATION 29
Final view of the application. 29
Explanation of the code. 30
WRITEUP A REAL-LIFE PROBLEM WHERE WE CAN APPLY AR 31
Problem Identification. 31
Solution 31
Benefits 31

INTRODUCTION

Augmented reality (AR) is an interactive experience of a real-world environment where the objects that reside in the real world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory.

- Wikipedia

We have been asked to create an Augmented reality application based on Tourism. In this report we implemented 4 different artifacts for display on the screen.

ASSIGNMENT TASKS

Task 1 – Create an AR application to display an artifact.

Task 2 – combine all projects to a one single application.

Task 3 – that you should use different markers to display multiple objects in a single app.

Task 4 - Make a small writeup about a real-life problem where we can apply AR technology to solve it.

METHEDOLOGY

We are asked to implement an AR application for tourists plan their tour destinations. It named as “Traveller”. There we have decided to show 4 different types of artifacts for the Tourists who wants to travel around the world. There we planned to show an Eiffel Tower, Moai Statues, Maison Carrée and Giza Pyramid. We used AR to show tourism destinations a great and massive experience to the audience. It provides an attractive interface for the tourists to plan their tours by checking before reach the destinations.

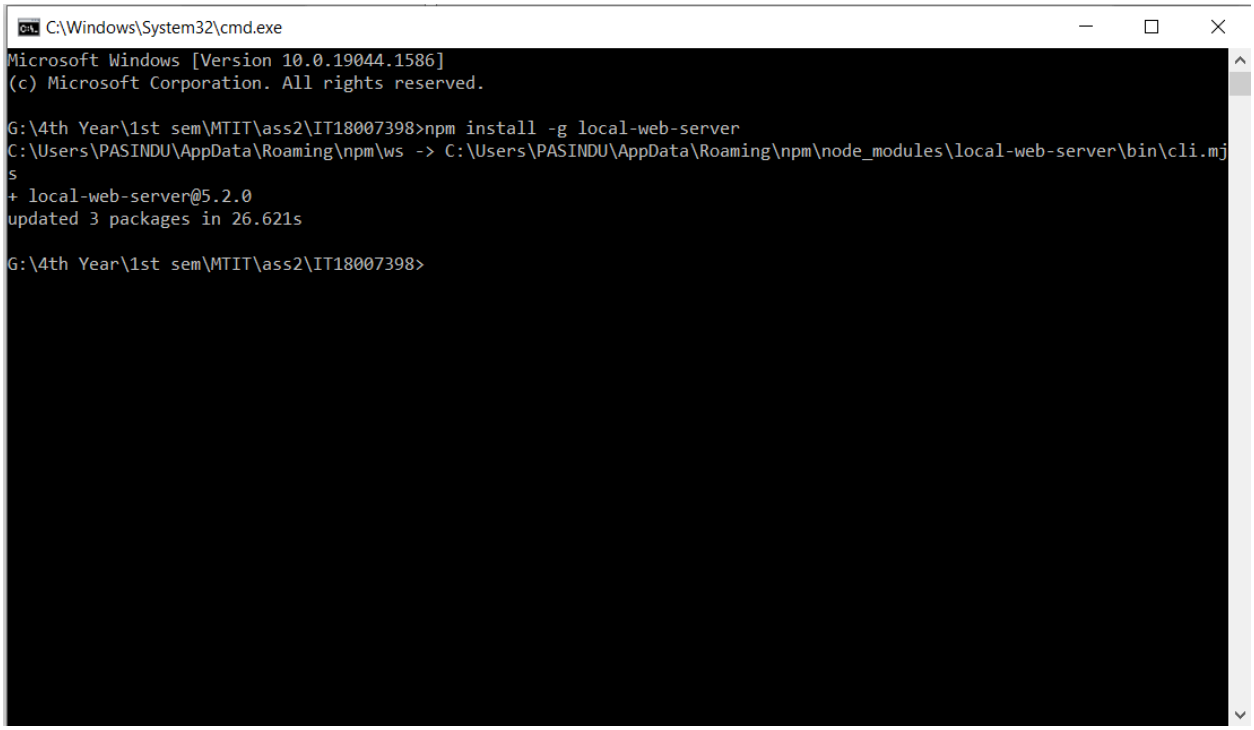
IMPLEMENTATION

Function 1- Eiffel Tower

Step 1 – Creating a secure server

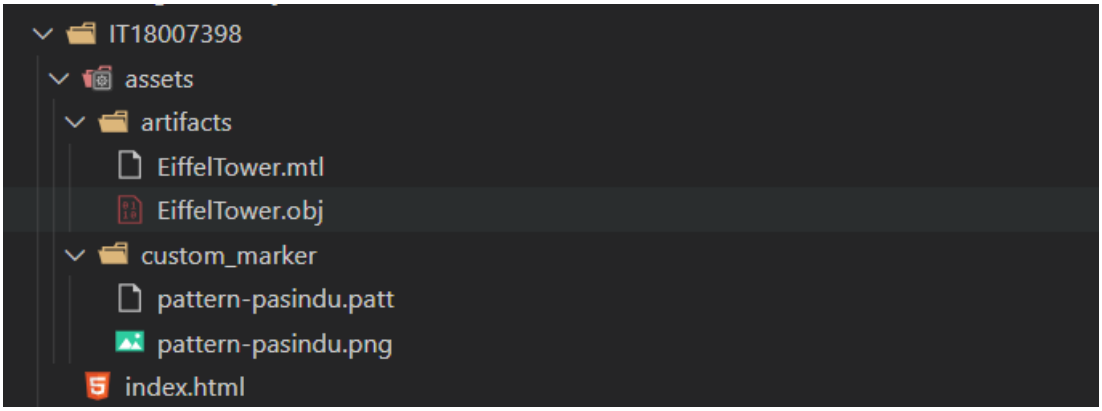
1. We need a secure server to host the application because it will access the camera of our mobile phone.
2. Therefore, open the terminal and install the bellow local web server.

   npm install -g local-web-server



Step 2 – Creating a project

1. Once crating the server, create a folder. This folder will be the project folder.
2. Create a text document and save it as HTML format.
3. Name the file as index.html.



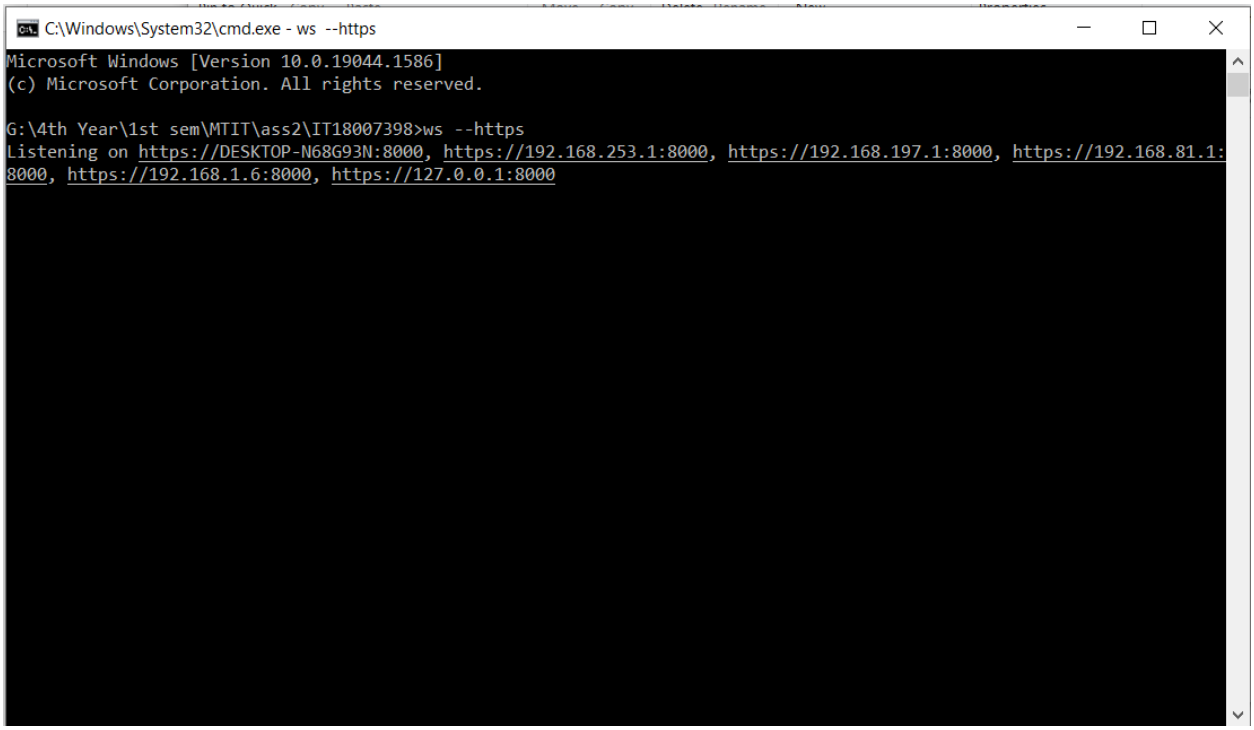
4. Insert the code as following.

```html
<!DOCTYPE html>
<html>

<head>
    <!-- include A-Frame obviously -->
    <script src="https://aframe.io/releases/0.6.0/aframe.min.js"></script>
    <!-- include ar.js for A-Frame -->
    <script src="https://jeromeetienne.github.io/AR.js/aframe/build/aframe-ar.js"></script>
</head>

<body style='margin : 0px; overflow: hidden;'>
    <a-scene embedded arjs>

        <a-assets>
            <a-asset-item id="etobj" src="assets/artifacts/EiffelTower.obj"></a-asset-item>
            <a-asset-item id="etmtl" src="assets/artifacts/EiffelTower.mtl"></a-asset-item>
        </a-assets>

        <a-marker preset='custom' type="pattern" url="assets/custom_marker/pattern-pasindu.patt">

            <a-entity obj-model="obj: #etobj;
                mtl: #etmtl " position="0 0.8 0" rotation="0 -45 0"
                scale="0.03 0.03 0.03">
            </a-entity>

            <a-entity position="0 0.3 1" text = "value: Eiffel Tower; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Pridarshana K.W.P.S.D IT18007398; color: white; wrap-count: 20"></a-entity>

        </a-marker>

        <a-entity camera></a-entity>

    </a-scene>
</body>

</html>
```

Step 3 – Run the server

1. Navigate to the project folder and open the terminal.
2. Run the following command

ws –https

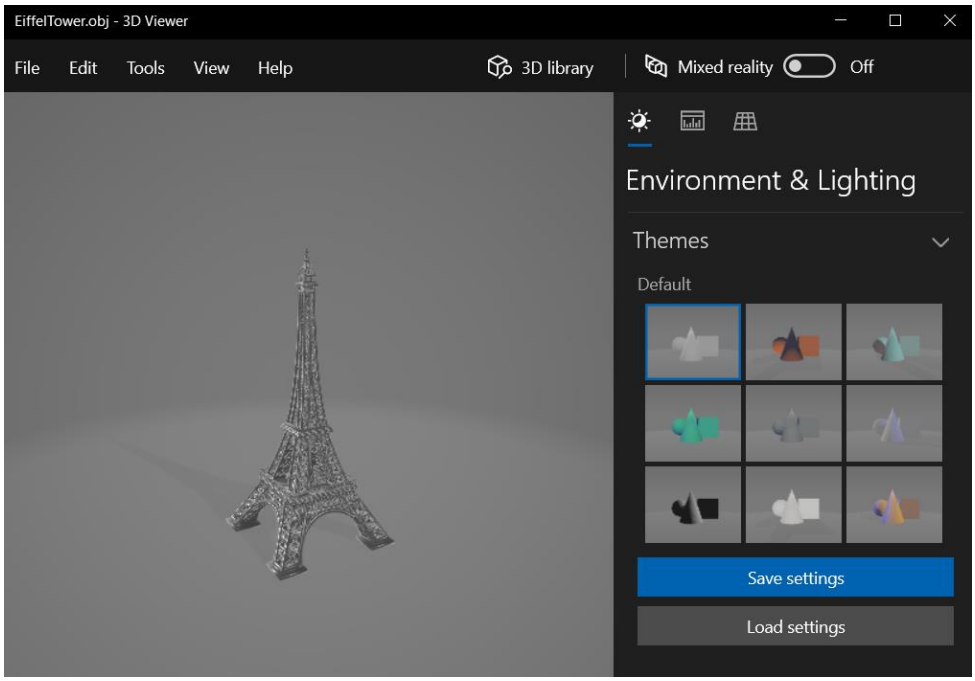


Step 4 – Open the browser and browse on the URL.

CODE EXPLAINATION

Step 1 – Select an artifact.

In the implementation I used 3D viewer to select an artifact. They provide numerous 3D objects. Selected 3D object is saved as GLTF binary format. The saved fie should be in project folder.



Step 2 – Select a Marker.

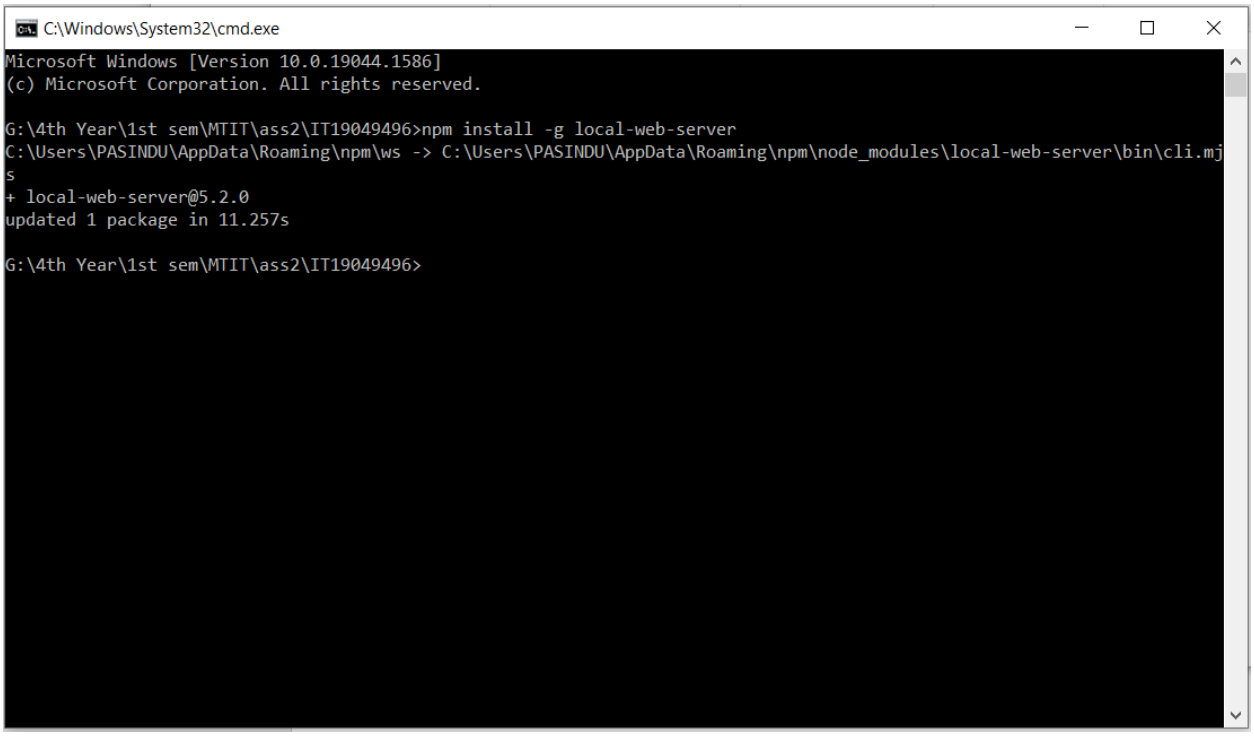
An image should be there for establishing the object on it. Therefore, an image is saved under the project folder and make it as a marker.

Function 2- Moai Statues

Step 1 – Creating a secure server

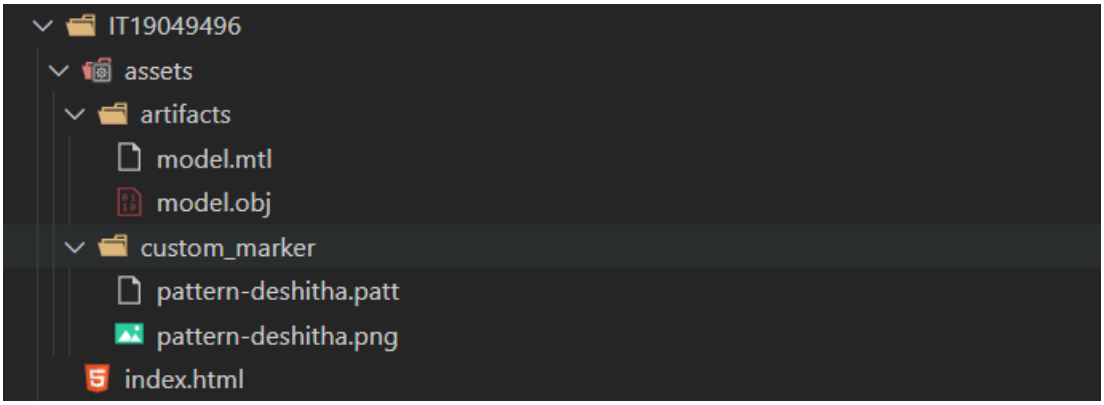
3. We need a secure server to host the application because it will access the camera of our mobile phone.
4. Therefore, open the terminal and install the bellow local web server.

   npm install -g local-web-server



Step 2 – Creating a project

5. Once crating the server, create a folder. This folder will be the project folder.
6. Create a text document and save it as HTML format.
7. Name the file as index.html.



8. Insert the code as following.

```html
<!DOCTYPE html>
<html>

<head>
    <!-- include A-Frame obviously -->
    <script src="https://aframe.io/releases/0.6.0/aframe.min.js"></script>
    <!-- include ar.js for A-Frame -->
    <script src="https://jeromeetienne.github.io/AR.js/aframe/build/aframe-ar.js"></script>
</head>

<body style='margin : 0px; overflow: hidden;'>
    <a-scene embedded arjs>

        <a-assets>
            <a-asset-item id="modelobj" src="assets/artifacts/model.obj"></a-asset-item>
            <a-asset-item id="modelmtl" src="assets/artifacts/model.mtl"></a-asset-item>
        </a-assets>

        <a-marker preset='custom' type="pattern" url="assets/custom_marker/pattern-deshitha.patt">

            <a-entity obj-model="obj: #modelobj;
                mtl: #modelmtl " position="0 0.8 0" rotation="-90 -45 0"
                scale="0.005 0.005 0.005">
            </a-entity>

            <a-entity position="0 0.3 1" text = "value: Moai; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Deshitha Thilindra H.B IT19049496; color: white; wrap-count: 18"></a-entity>

        </a-marker>

        <a-entity camera></a-entity>

    </a-scene>
</body>

</html>
```

Step 3 – Run the server

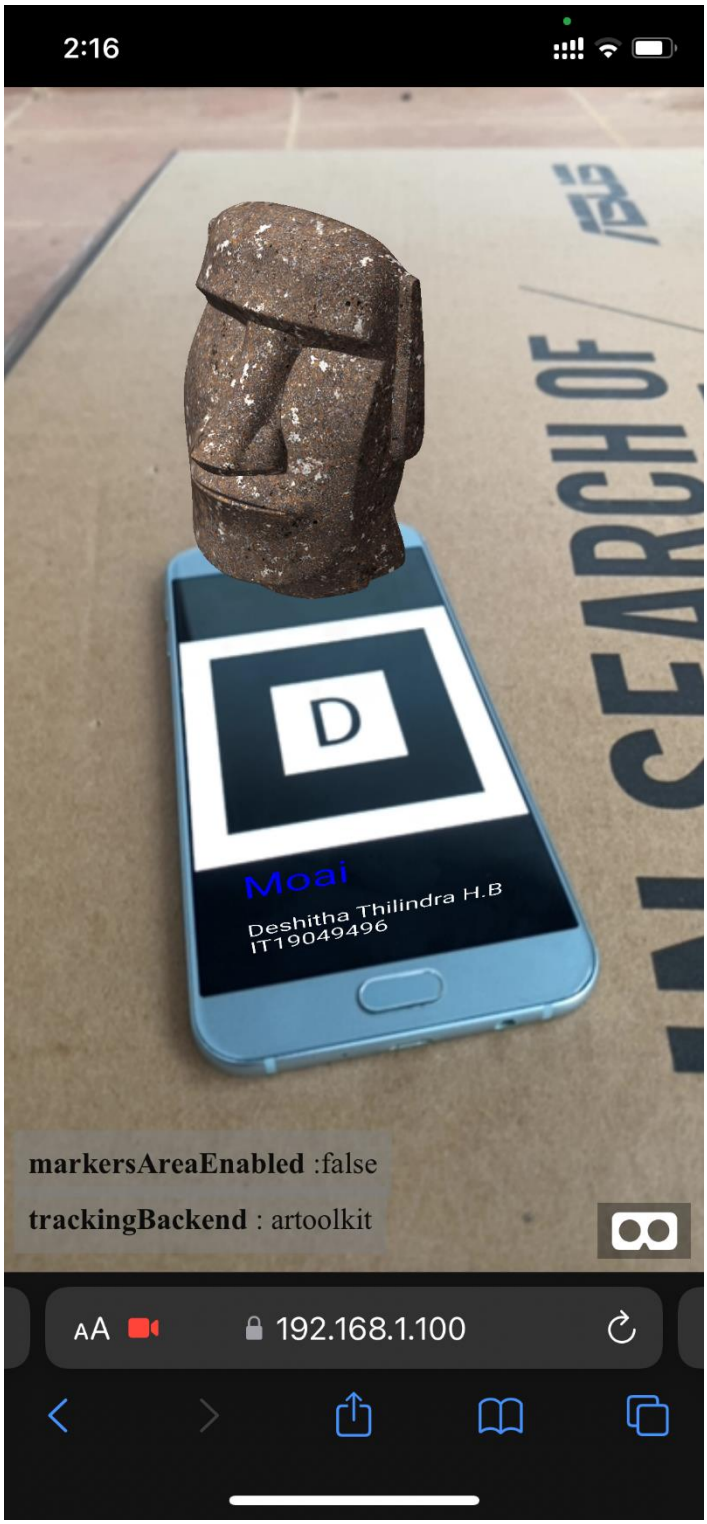
3. Navigate to the project folder and open the terminal.
4. Run the following command

ws –https

C:\Windows\System32\cmd.exe - ws --https

```
Microsoft Windows [Version 10.0.19044.1586]
(c) Microsoft Corporation. All rights reserved.

G:\4th Year\1st sem\MTIT\ass2\IT19049496>ws --https
Listening on https://DESKTOP-N68G93N:8000, https://192.168.253.1:8000, https://192.168.197.1:8000, https://192.168.81.1:
8000, https://192.168.1.6:8000, https://127.0.0.1:8000
```

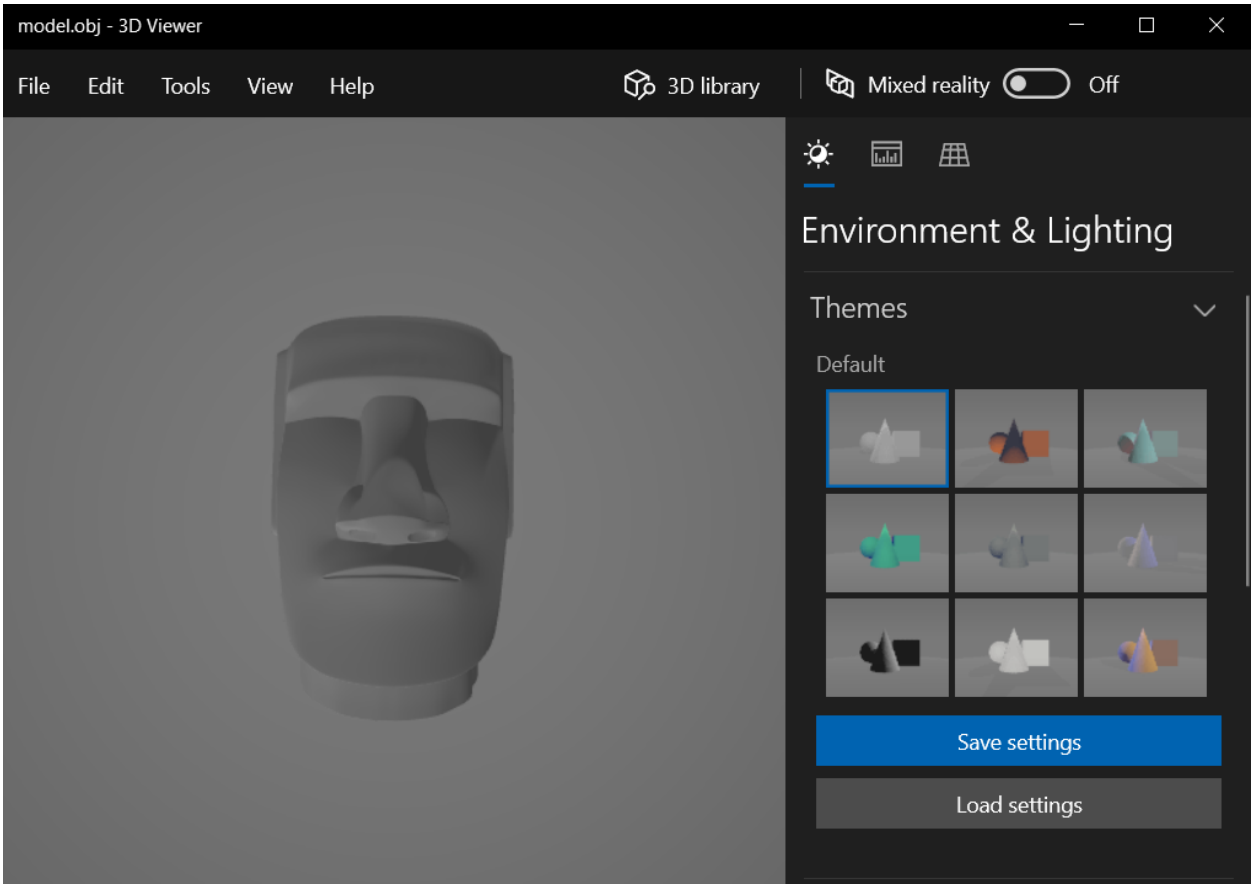
Step 4 – Open the browser and browse on the URL.



CODE EXPLAINATION

Step 1 – Select an artifact.

In the implementation I used 3D viewer to select an artifact. They provide numerous 3D objects. Selected 3D object is saved as GLTF binary format. The saved fie should be in project folder.



Step 2 – Select a Marker.

An image should be there for establishing the object on it. Therefore, an image is saved under the project folder and make it as a marker.

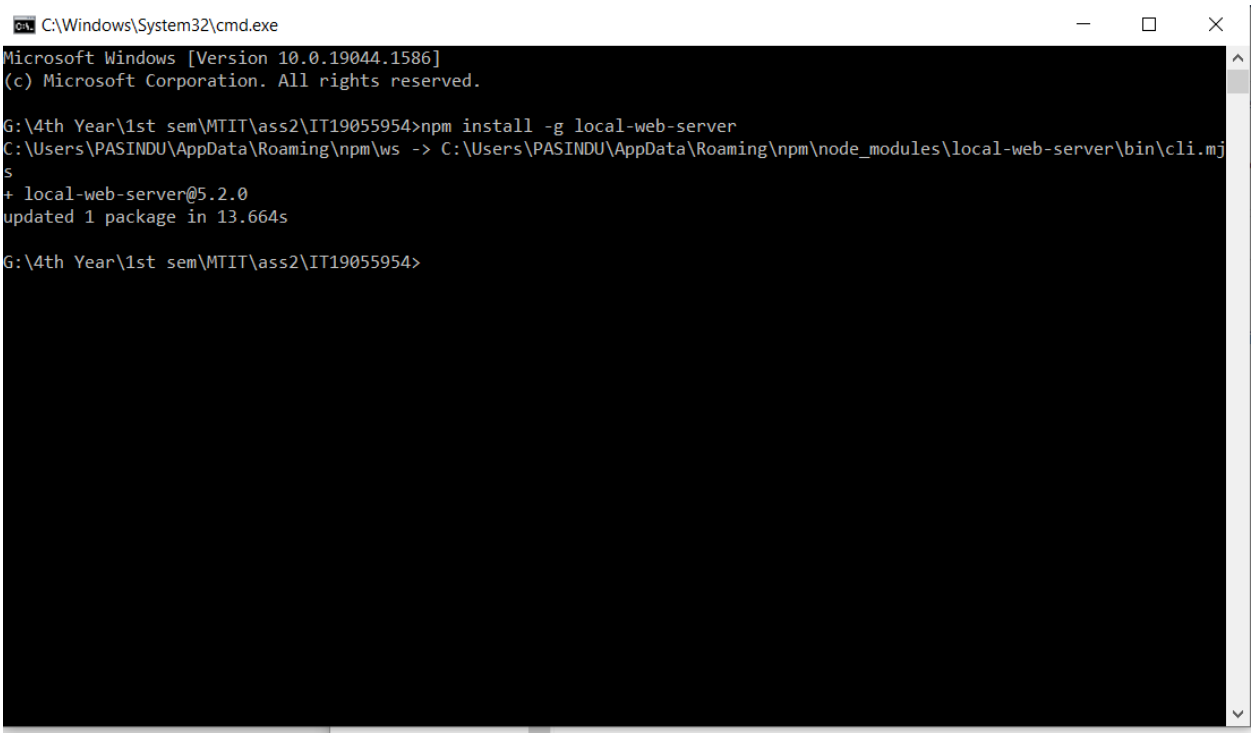
D

Function 3- Maison Carrée

Step 1 – Creating a secure server

5. We need a secure server to host the application because it will access the camera of our mobile phone.
6. Therefore, open the terminal and install the bellow local web server.

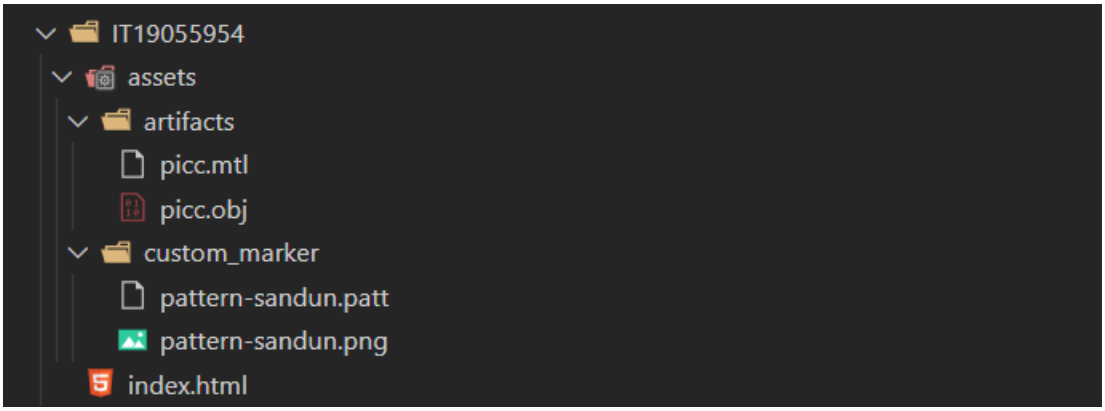
   npm install -g local-web-server



Step 2 – Creating a project

9. Once crating the server, create a folder. This folder will be the project folder.
10. Create a text document and save it as HTML format.

11. Name the file as index.html.



12. Insert the code as following.

```
<!DOCTYPE html>
<html>

<head>
    <!-- include A-Frame obviously -->
    <script src="https://aframe.io/releases/0.6.0/aframe.min.js"></script>
    <!-- include ar.js for A-Frame -->
    <script src="https://jeromeetienne.github.io/AR.js/aframe/build/aframe-ar.js"></script>
</head>

<body style='margin : 0px; overflow: hidden;'>
    <a-scene embedded arjs>

        <a-assets>
            <a-asset-item id="3dobj" src="assets/artifacts/picc.obj"></a-asset-item>
            <a-asset-item id="3dmtl" src="assets/artifacts/picc.mtl"></a-asset-item>
        </a-assets>

        <a-marker preset='custom' type="pattern" url="assets/custom_marker/pattern-sandun.patt">

            <a-entity obj-model="obj: #3dobj;
                mtl: #3dmtl " position="0 0.8 0" rotation="0 45 0"
                scale="0.00038 0.00038 0.00038">
            </a-entity>

            <a-entity position="0 0.3 1" text = "value: Maison Carrée; color: blue; wrap-count: 12"></a-entity>
            <a-entity position="0 0 1" text = "value: Sandun Samaranayake S.G  IT19055954; color: white; wrap-count: 22"></a-entity>

        </a-marker>

        <a-entity camera></a-entity>

    </a-scene>
</body>

</html>
```

Step 3 – Run the server

5. Navigate to the project folder and open the terminal.
6. Run the following command

ws –https

C:\Windows\System32\cmd.exe - ws --https

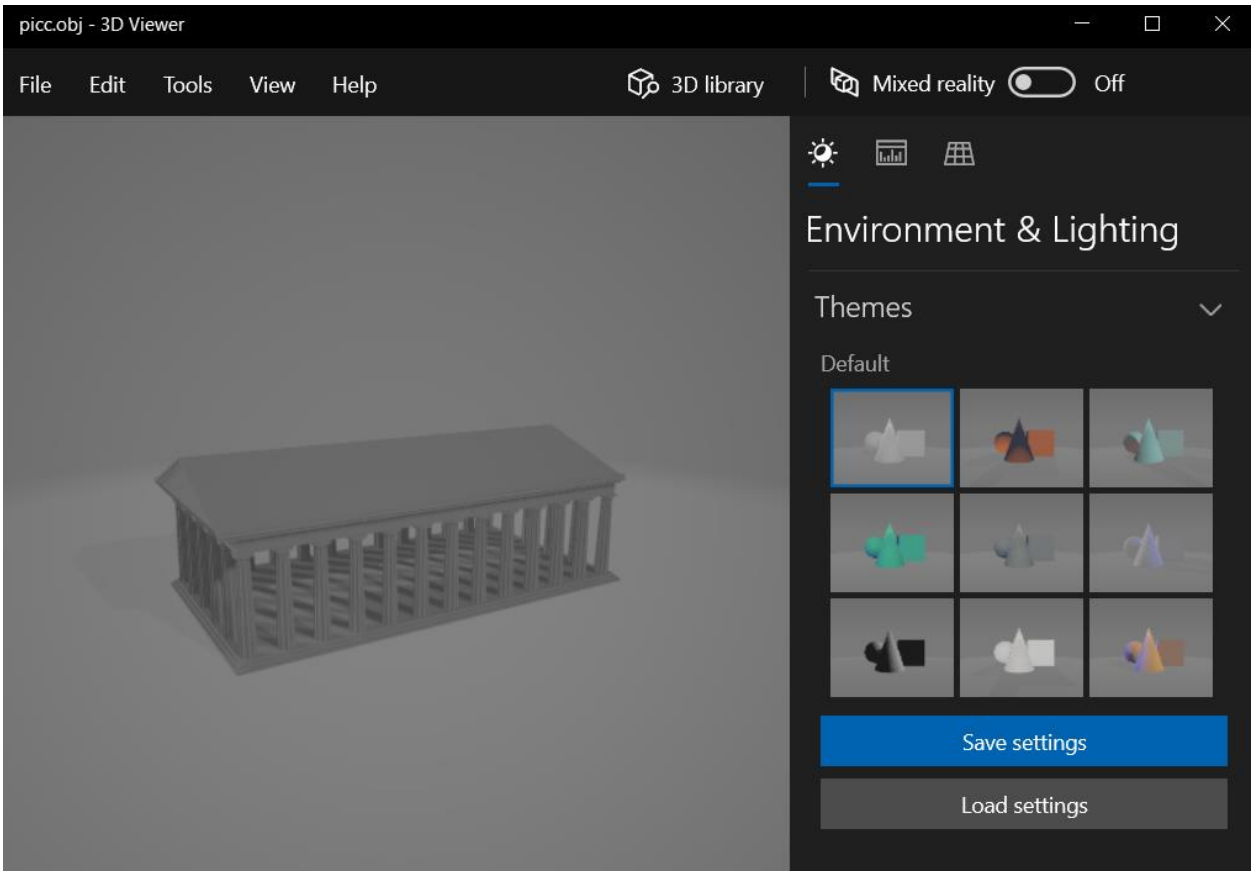
```
Microsoft Windows [Version 10.0.19044.1586]
(c) Microsoft Corporation. All rights reserved.

G:\4th Year\1st sem\MTIT\ass2\IT19055954>ws --https
Listening on https://DESKTOP-N68G93N:8000, https://192.168.253.1:8000, https://192.168.197.1:8000, https://192.168.81.1:
8000, https://192.168.1.6:8000, https://127.0.0.1:8000
```

Step 4 – Open the browser and browse on the URL.

CODE EXPLAINATION

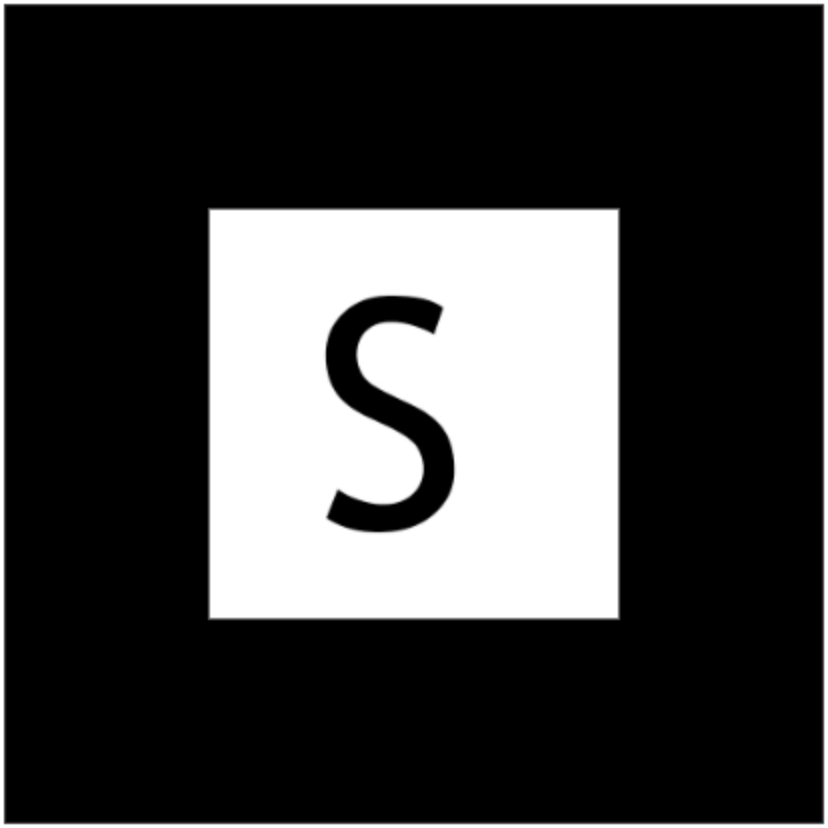
Step 1 – Select an artifact.

In the implementation I used 3D viewer to select an artifact. They provide numerous 3D objects. Selected 3D object is saved as GLTF binary format. The saved fie should be in project folder.



Step 2 – Select a Marker.

An image should be there for establishing the object on it. Therefore, an image is saved under the project folder and make it as a marker.

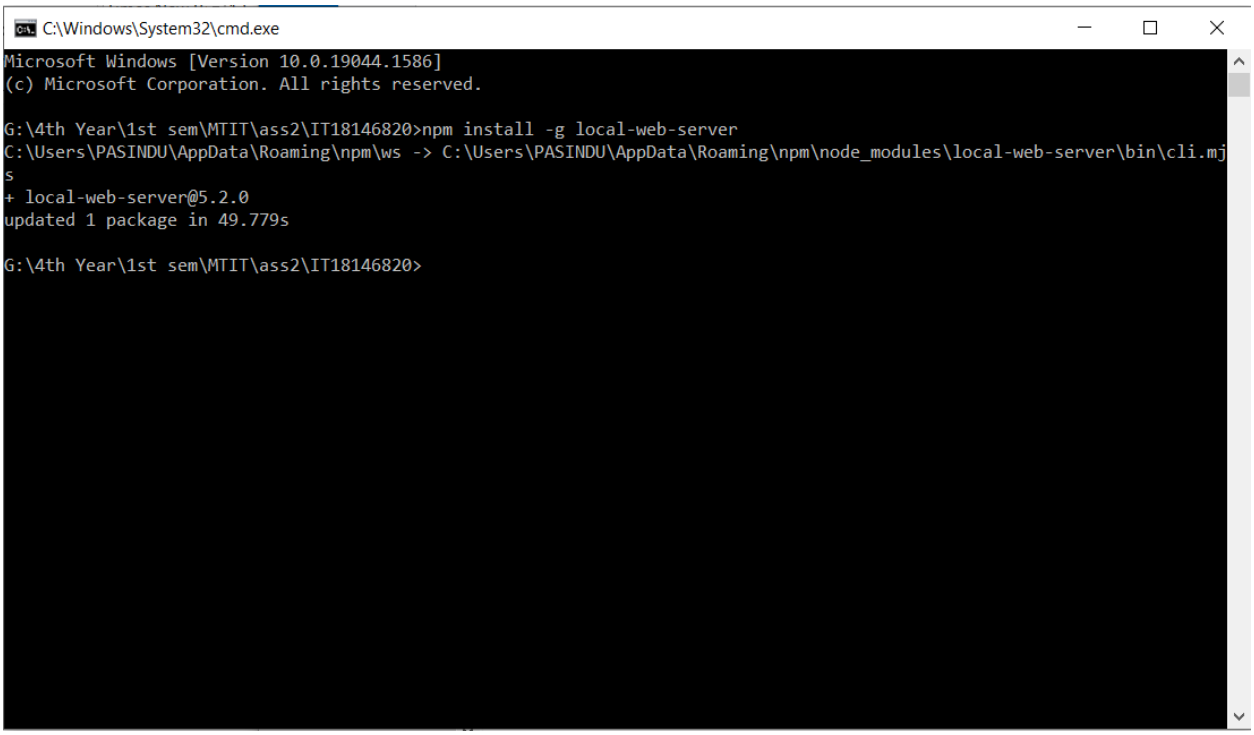


Function 4 – Giza Pyramid

Step 1 – Creating a secure server

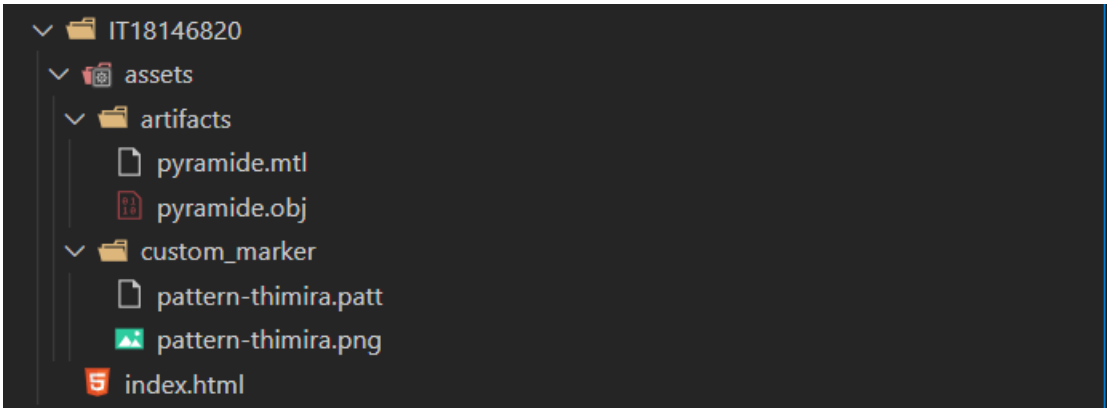
7. We need a secure server to host the application because it will access the camera of our mobile phone.
8. Therefore, open the terminal and install the bellow local web server.

   npm install -g local-web-server



Step 2 – Creating a project

13. Once crating the server, create a folder. This folder will be the project folder.
14. Create a text document and save it as HTML format. Name the file as index.html.



15. Insert the code as following.

```html
<!DOCTYPE html>
<html>

<head>
    <!-- include A-Frame obviously -->
    <script src="https://aframe.io/releases/0.6.0/aframe.min.js"></script>
    <!-- include ar.js for A-Frame -->
    <script src="https://jeromeetienne.github.io/AR.js/aframe/build/aframe-ar.js"></script>
</head>

<body style='margin : 0px; overflow: hidden;'>
    <a-scene embedded arjs>

        <a-assets>
            <a-asset-item id="prmdobj" src="assets/artifacts/pyramide.obj"></a-asset-item>
            <a-asset-item id="prmdmtl" src="assets/artifacts/pyramide.mtl"></a-asset-item>
        </a-assets>

        <a-marker preset='custom' type="pattern" url="assets/custom_marker/pattern-thimira.patt">

            <a-entity obj-model="obj: #prmdobj;
                mtl: #prmdmtl " position="0 0.8 0" rotation="0 -45 0"
                scale="4 4 4">
            </a-entity>

            <a-entity position="0 0.3 1" text = "value: Pyramide; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Kavinda K.T.S IT18146820; color: white; wrap-count: 15"></a-entity>

        </a-marker>

        <a-entity camera></a-entity>

    </a-scene>
</body>

</html>
```

Step 3 – Run the server

7. Navigate to the project folder and open the terminal.
8. Run the following command

ws –https

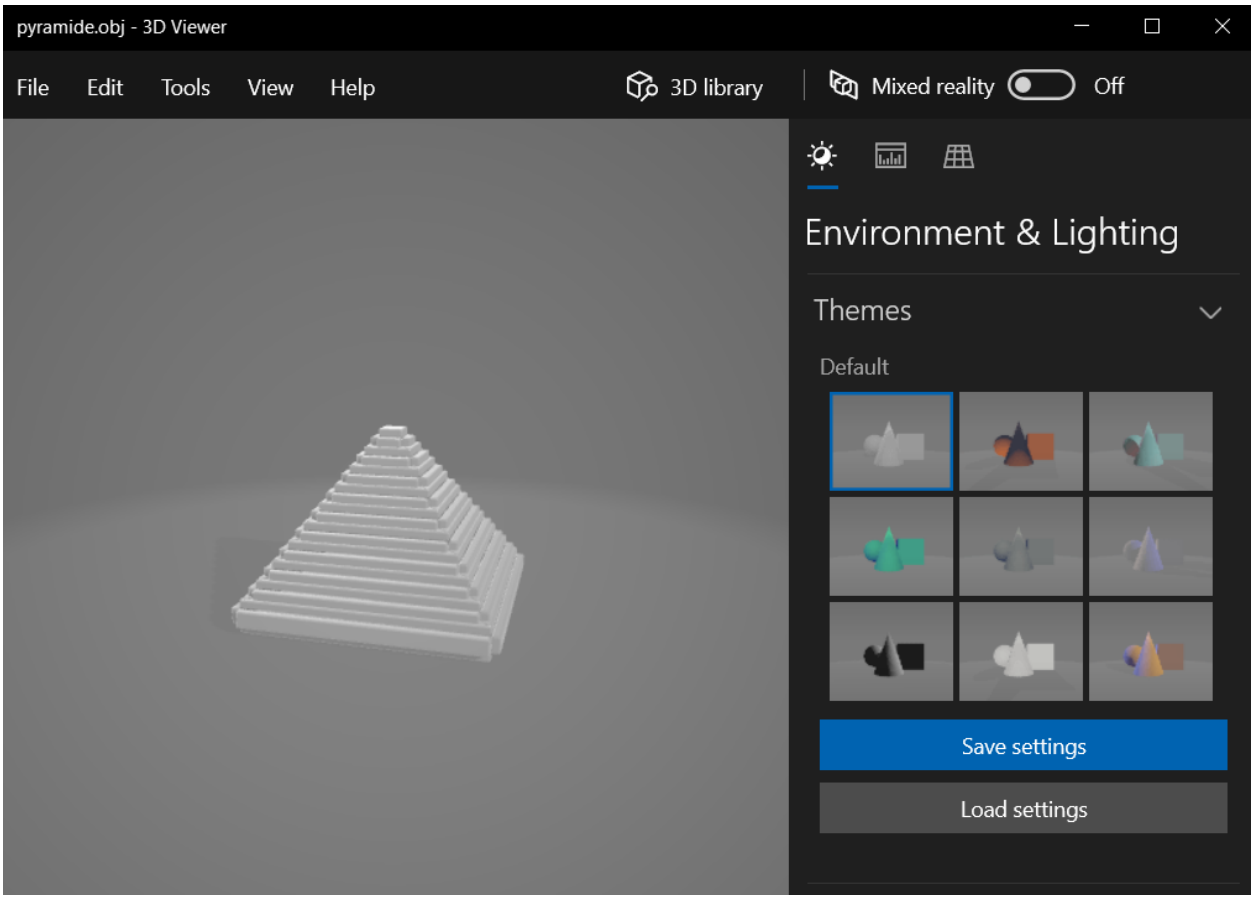
```
D:\SLIIT\SEMS\4th yr\MTIT\Final\2\IT18146820>npm install -g local-web-server

changed 188 packages, and audited 189 packages in 13s

11 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\SLIIT\SEMS\4th yr\MTIT\Final\2\IT18146820>ws --https
Listening on https://ddthilindra:8000, https://192.168.1.102:8000, https://127.0.0.1:8000
```

Step 4 – Open the browser and browse on the URL.

CODE EXPLAINATION

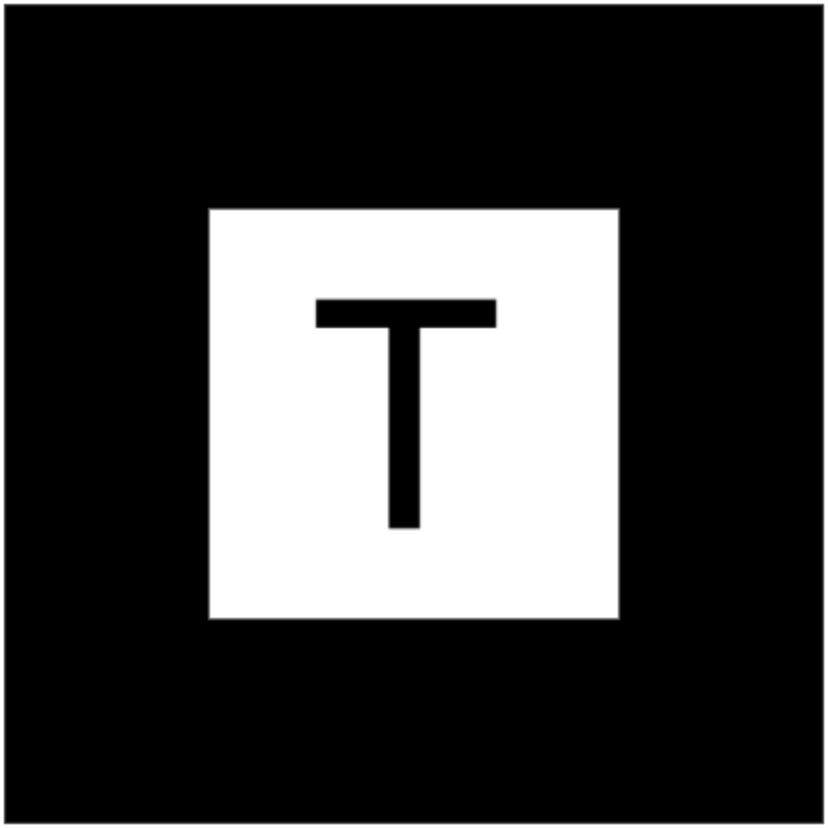
Step 1 – Select an artifact.

In the implementation I used 3D viewer to select an artifact. They provide numerous 3D objects. Selected 3D object is saved as GLB binary format. The saved fie should be in project folder.



Step 2 – Select a Marker.

An image should be there for establishing the object on it. Therefore, an image is saved under the project folder and make it as a marker.

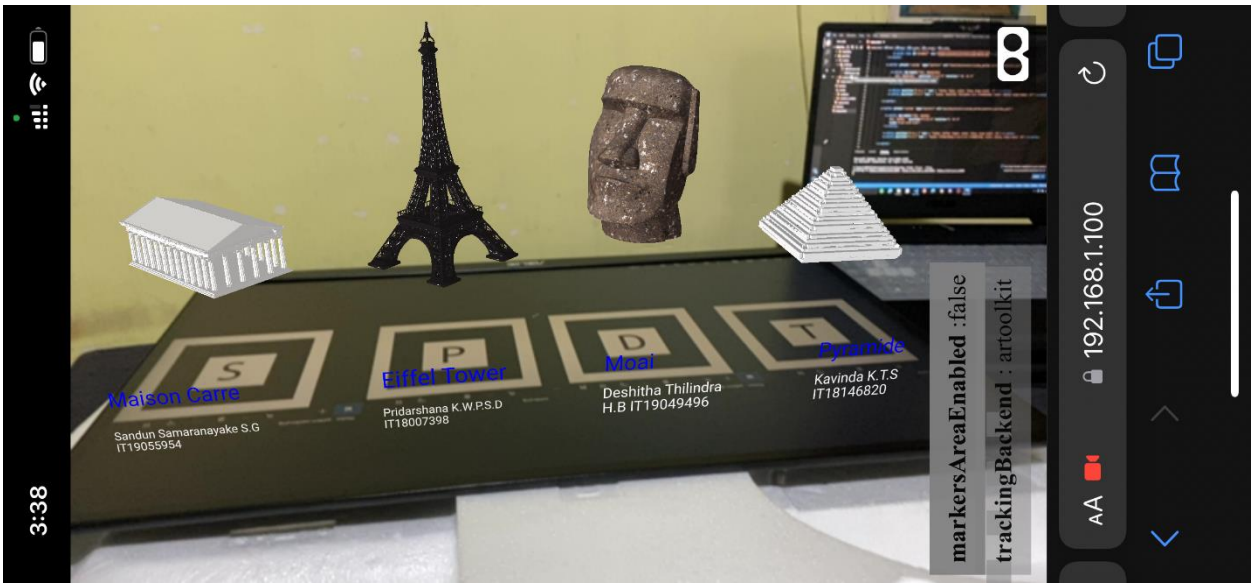


FINAL APPLICATION

Final view of the application.

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <link rel="stylesheet" href="home.css" media="all" />
    <title>Traveller | AR Assignment</title>
  </head>
  <body>
    <script>
      function sandun() {
        location.replace("/IT19055954/");
      }
      function deshitha() {
        location.replace("/IT19049496/");
      }
      function pasindu() {
        location.replace("/IT18007398/");
      }
      function thimira() {
        location.replace("/IT18146820/");
      }
      function Integrated() {
        location.replace("/Integrated Project/");
      }
    </script>
    <div class="container">
      <h1>Traveller</h1>
      <button onclick="sandun()">
        <section class="cards">
          <article class="card card--1">
            <div class="card__img"></div>
            <a href="#" class="card_link">
              <div class="card__img--hover"></div>
            </a>
            <div class="card__info">
              <h3 class="card__title">Maison Carree</h3>
              <span class="card__by"
                >Created by
                <a href="#" class="card__author" title="author"
                  >Sandun Samaranayake S.G  IT19055954</a
                ></span>
            </div>
          </article>
        </section>
      </button>

      <button onclick="deshitha()">
        <section class="cards">
          <article class="card card--2">
            <div class="card__img"></div>
            <a href="#" class="card_link">
              <div class="card__img--hover"></div>
            </a>
            <div class="card__info">
              <h3 class="card__title">Moai</h3>
              <span class="card__by"
                >Created by
                <a href="#" class="card__author" title="author"
                  >Deshitha Thilindra H.B IT19049496</a
                ></span>
            </div>
          </article>
        </section>
      </button>

      <button onclick="pasindu()">
        <section class="cards">
          <article class="card card--3">
            <div class="card__img"></div>
            <a href="#" class="card_link">
              <div class="card__img--hover"></div>
            </a>
            <div class="card__info">
              <h3 class="card__title">Eiffel Tower</h3>
              <span class="card__by"
                >Created by
                <a href="#" class="card__author" title="author"
                  >Pridarshana K.W.P.S.D IT18007398</a
                ></span>
            </div>
          </article>
        </section>
      </button>

      <button onclick="thimira()">
        <section class="cards">
          <article class="card card--4">
            <div class="card__img"></div>
            <a href="#" class="card_link">
              <div class="card__img--hover"></div>
            </a>
            <div class="card__info">
              <h3 class="card__title">Pyramide</h3>
              <span class="card__by"
                >Created by
                <a href="#" class="card__author" title="author"
                  >Kavinda K.T.S IT18146820</a
                ></span>
            </div>
          </article>
        </section>
      </button>

      <button onclick="Integrated()">
        <section class="cards">
          <article class="card card--5">
            <div class="card__img"></div>
            <a href="#" class="card_link">
              <div class="card__img--hover"></div>
            </a>
            <div class="card__info">
              <h3 class="card__title">Integrated Project</h3>
            </div>
          </article>
        </section>
      </button>
    </div>
  </body>
</html>
```

Explanation of the code.




```
<!DOCTYPE html>
<html>
<head>
    <meta charset='utf-8'>
    <meta http-equiv='X-UA-Compatible' content='IE=edge'>
    <title>AR Assignment</title>
    <!-- include A-Frame obviously -->
    <script src="https://aframe.io/releases/0.6.0/aframe.min.js"></script>
    <!-- include ar.js for A-Frame -->
    <script src="https://jeromeetienne.github.io/AR.js/aframe/build/aframe-ar.js"></script>
</head>
<body style='margin : 0px; overflow: hidden;'>
    <a-scene embedded arjs>

        <a-assets>
            <a-asset-item id="modelobj" src="../IT19049496/assets/artifacts/model.obj"></a-asset-item>
            <a-asset-item id="modelmtl" src="../IT19049496/assets/artifacts/model.mtl"></a-asset-item>

            <a-asset-item id="etobj" src="../IT18007398/assets/artifacts/EiffelTower.obj"></a-asset-item>
            <a-asset-item id="etmtl" src="../IT18007398/assets/artifacts/EiffelTower.mtl"></a-asset-item>

            <a-asset-item id="3dobj" src="../IT19055954/assets/artifacts/picc.obj"></a-asset-item>
            <a-asset-item id="3dmtl" src="../IT19055954/assets/artifacts/picc.mtl"></a-asset-item>

            <a-asset-item id="prmdobj" src="../IT18146820/assets/artifacts/pyramide.obj"></a-asset-item>
            <a-asset-item id="prmdmtl" src="../IT18146820/assets/artifacts/pyramide.mtl"></a-asset-item>
        </a-assets>

        <a-marker preset='custom' type="pattern" url="../IT19049496/assets/custom_marker/pattern-deshitha.patt">

            <a-entity obj-model="obj: #modelobj; mtl: #modelmtl " position="0 0.8 0" rotation="-90 -45 0" scale="0.005 0.005 0.005"></a-entity>

            <a-entity position="0 0.3 1" text = "value: Moai; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Deshitha Thilindra H.B IT19049496; color: white; wrap-count: 18"></a-entity>

        </a-marker>
```

```
        <a-marker preset='custom' type="pattern" url="../IT18007398/assets/custom_marker/pattern-pasindu.patt">

            <a-entity obj-model="obj: #etobj; mtl: #etmtl " position="0 0.8 0" rotation="0 -45 0" scale="0.03 0.03 0.03"></a-entity>

            <a-entity position="0 0.3 1" text = "value: Eiffel Tower; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Pridarshana K.W.P.S.D IT18007398; color: white; wrap-count: 20"></a-entity>

        </a-marker>

        <a-marker preset='custom' type="pattern" url="../IT19055954/assets/custom_marker/pattern-sandun.patt">

            <a-entity obj-model="obj: #3dobj; mtl: #3dmtl " position="0 0.8 0" rotation="0 45 0"scale="0.00038 0.00038 0.00038"></a-entity>

            <a-entity position="0 0.3 1" text = "value: Maison Carrée; color: blue; wrap-count: 12"></a-entity>
            <a-entity position="0 0 1" text = "value: Sandun Samaranayake S.G IT19055954; color: white; wrap-count: 22"></a-entity>

        </a-marker>

        <a-marker preset='custom' type="pattern" url="../IT18146820/assets/custom_marker/pattern-thimira.patt">

            <a-entity obj-model="obj: #prmdobj; mtl: #prmdmtl " position="0 0.8 0" rotation="0 -45 0" scale="4 4 4"></a-entity>

            <a-entity position="0 0.3 1" text = "value: Pyramide; color: blue; wrap-count: 10"></a-entity>
            <a-entity position="0 0 1" text = "value: Kavinda K.T.S IT18146820; color: white; wrap-count: 15"></a-entity>

        </a-marker>

        <a-entity camera></a-entity>

    </a-scene>
</body>

</html>
```

WRITEUP A REAL-LIFE PROBLEM WHERE WE CAN APPLY AR

Problem Identification.

Problems arise when tourists travel around the world with little or no understanding of where they are traveling. Then they have to enlist the help of others. Then they have to pay more.

Solution

AR Technology allows tourists to get an idea of their destination before they reach it and to explore those places.

Benefits

- Improved Information:
    - Augmented reality provides additional details on the Places.
- Easy Visualization:
    - AR makes it possible to visualize an object in a personalized context.